BASE CAMP · IGCSE MATHEMATICS · GRADE 7 · NUMBER

# Fractions, Decimals and Percentages — Paper D

HARD · NON-CALCULATOR · 50 MINUTES · 30 MARKS

NAME DATE SCORE / 30

- Answer every question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- No calculator. Anything you cannot do in your head, do on paper.
- Give fractions in their simplest form unless the question says otherwise.
- The mark for each question is shown in square brackets on the right.
- Two questions on this paper have no obvious first step. If one stops you, leave it and come back — do not spend more than five minutes stuck.

SECTION 1 — PROFIT AND LOSS

**1.** A trader buys an item for 480 rupees and sells it for 600 rupees. Calculate the percentage profit. [3]

**2.** A shopkeeper sells a bicycle at a 20% loss, receiving 640 rupees for it. Find the price he originally paid. [3]

*A 20% loss means he received 80% of what he paid. That is the same shape of problem as a reverse percentage.*

SECTION 2 — FRACTIONS OF WHAT IS LEFT

**3.** Anil spends 1/3 of his money on books, and then 1/4 of what remains on food. He is left with 900 rupees.

**(a)** What fraction of his original money does he have left? [2]

**(b)** How much money did he start with? [2]

**4.** A tank is 3/5 full. After 240 litres are removed, it is 1/3 full. Find the total capacity of the tank. [4]

SECTION 3 — COMPARISONS AND TRAPS

**5.** A shop offers two deals on an item priced at 600 rupees. [4]
Deal 1: 20% off, then a further 10% off.
Deal 2: 25% off, then a further 5% off.
Work out the final price under each deal, and state which is better and by how much.

**6.** The price of an item rises by 25%. By what percentage must the new price now fall to bring it back to exactly the original price? [4]

*The answer is not 25%. Work with a starting value of 100 if that helps you see it.*

**7.** In a school of 500 students, 3/5 are girls. 25% of the girls and 40% of the boys walk to school. How many students walk to school in total? [4]

**8.** A student has written two answers in her book, and both are wrong. [4]
(i) “0.9 is smaller than 0.15, because 15 is bigger than 9.”
(ii) “1/2 + 1/3 = 2/5”
For each one, explain the mistake and give the correct answer.

*End of paper. Check every answer has a unit or a form the question actually asked for.*